



# **Video Games in Steam: Capital Market & Female Fandom Community**

Jingjing Wang

Celia Liang

# Abstract

Made by Apache Spark

- Research Context & Motivation
- Methodology
- Technical Approach
- Key Objective

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# Motivation

## Our Original Intention:

As female gamers, we find out the commercial success of video games and its female community reputation do not always align. A game can dominate sales charts while offering little meaningful female representation, and conversely, games with strong female fanbases on AO3 may not rank among top commercial performers.

By mapping these two dimensions across a range of game titles and genres, we aim to surface patterns that reveal the scale of an underserved market.

## People who might benefit:

1. Female players seeking games suited to their tastes.
2. Female players who wish to be seen (e.g., in the face of claims by male players that games cater exclusively to men).
3. Game developers who strive to accommodate the needs of audiences of all genders.

# Goodness

## 1. Data Validity

- Multi-Source Integration
- Scientific Grounding
- Temporal Alignment

## 2. Methodological Validity

- Correlation Analysis
- Precedent Framework
- Beyond Sales Figures

## 3: Robustness & Reproducibility

- Temporal Sensitivity Testing
- Multi-Dimensional Cross-Validation
- Control for Confounding Factors
- Standardized Workflow

# Data Sources

## Dataset 1

**Name:** AO3 Datasets — Selective data dump for fan statisticians

**Description:** Including two CSV files. The first one stores information about the **works**, the second one provides the key to the **tag IDs**.

**Size of data:** 417 MB

**Link to data:**

[https://archiveofourown.org/admin\\_posts/18804](https://archiveofourown.org/admin_posts/18804)

## Dataset 2

**Name:** Steamspy Dataset — Basic info of top 10000 popular games

**Description:** Data scraped from Steam using the Steamspy API, include information about download count, sale price, ccu, etc.

**Size of data:** 3.7MB

**Link to data:**

/Users/sindawjj/PycharmProjects/PBDworm/s  
teamspy\_top10000\_raw.jsonl

# Data Sample — Dataset 1

## Works

| creation date | language | complete | word_count | tags                      |
|---------------|----------|----------|------------|---------------------------|
| 26/2/21       | en       | TRUE     | 388        | 10+414093+1001939+4577144 |
| 26/2/21       | zh       | TRUE     | 1638       | 10+20350917+34816907+...  |
| 26/2/21       | en       | FALSE    | 3016       | 11+209651+37148719+...    |

## Tags

| id     | type      | name           | canonical | cached_count |
|--------|-----------|----------------|-----------|--------------|
| 414093 | Fandom    | Supernatural   | TRUE      | 310,300      |
| 27     | Fandom    | Genshin Impact | TRUE      | 85,000       |
| 25     | Character | Sam Winchester | TRUE      | 155,832      |

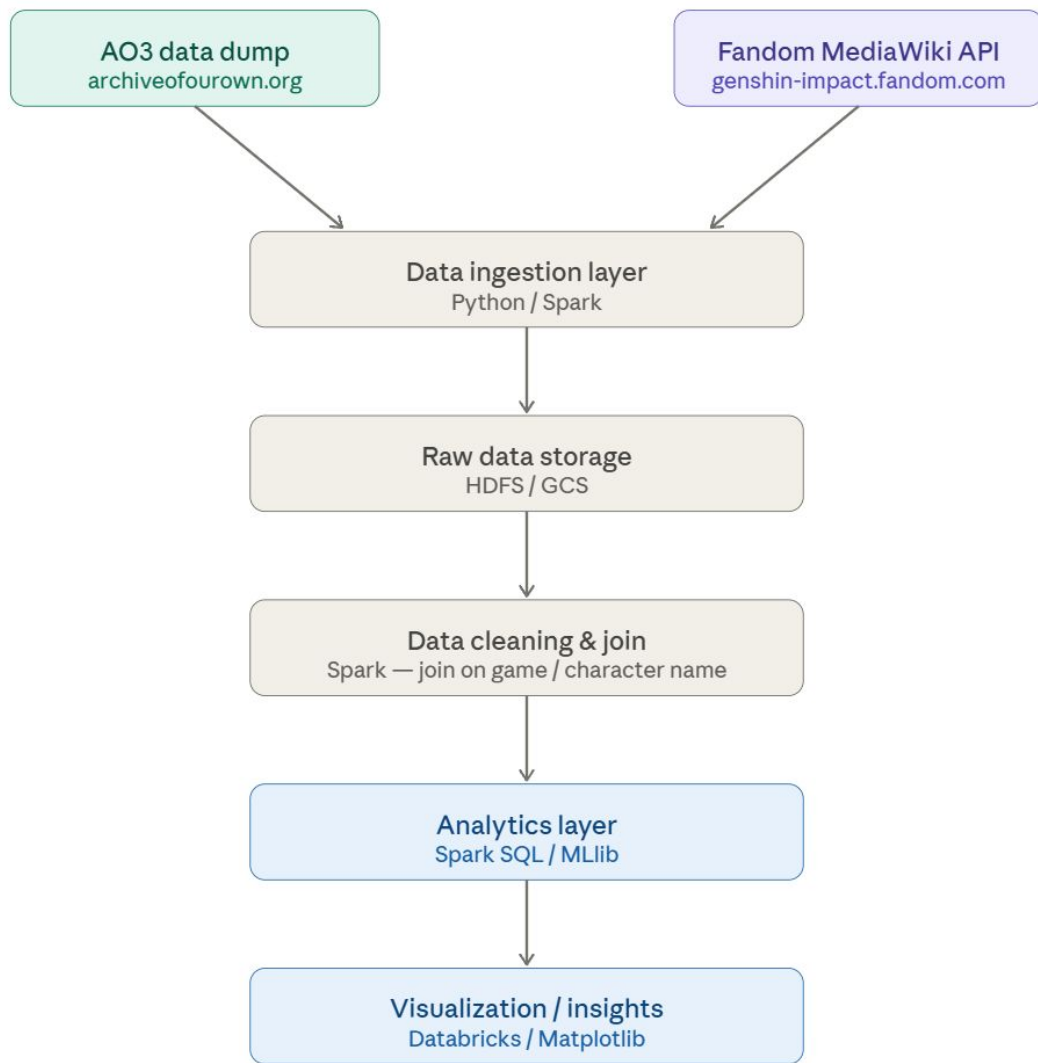
# Data Sample — Dataset 2

```
1 {"appid": 730, "name": "Counter-Strike: Global Offensive", "developer": "Valve", "publisher": "Valve", "score_rank": 1} ✓
2 {"appid": 1172470, "name": "Apex Legends", "developer": "Respawn", "publisher": "Electronic Arts", "score_rank": 2}
3 {"appid": 578080, "name": "PUBG: BATTLEGROUNDS", "developer": "PUBG Corporation", "publisher": "KRAFTON", "score_rank": 3}
4 {"appid": 1623730, "name": "Palworld", "developer": "Pocketpair", "publisher": "Pocketpair", "score_rank": 4}
5 {"appid": 440, "name": "Team Fortress 2", "developer": "Valve", "publisher": "Valve", "score_rank": 5}
6 {"appid": 1938090, "name": "Call of Duty: Modern Warfare II", "developer": "Treyarch, Raven Software", "publisher": "Activision", "score_rank": 6}
7 {"appid": 1063730, "name": "New World: Aeternum", "developer": "Amazon Games", "publisher": "Amazon Games", "score_rank": 7}
8 {"appid": 2358720, "name": "Black Myth: Wukong", "developer": "Game Science", "publisher": "Game Science", "score_rank": 8}
9 {"appid": 271590, "name": "Grand Theft Auto V Legacy", "developer": "Rockstar North", "publisher": "Rockstar Games", "score_rank": 9}
10 {"appid": 550, "name": "Left 4 Dead 2", "developer": "Valve", "publisher": "Valve", "score_rank": 10}
11 {"appid": 304930, "name": "Unturned", "developer": "Smartly Dressed Games", "publisher": "Smartly Dressed Games", "score_rank": 11}
```

## Columns:

appid, name, developer, publisher, owners, average\_forever, average\_2weeks, median\_forever, median\_2weeks, ccu, price, initialprice, discount, negative, positive, userscore (null)

# Design Diagram





# Code Challenge

## Problem

When loading a script in the Spark Shell using `:load`, if a `.withColumn` call is split across lines—specifically when the preceding line lacks a trailing operator—the Scala interpreter may sometimes interpret the first line as a complete statement.

```
val dfPrice = dfNotNull.withColumn(...)
```

```
.withColumn(...) // This might be interpreted as a new line of code
```

Result: `dfPrice` successfully incorporated only the first `.withColumn`; the second `.withColumn` became an anonymous operation, and Spark Shell created a temporary variable—`res6`—to store the second column.

## Solution

Avoid splitting lines with leaving a trailing operator, write code like:

```
val dfPrice = dfNotNull.withColumn(...) .withColumn(...
```

```
...)
```

# Code Challenge

## Problem

Exact matching (===) on game names yielded only ~300 matches

Naming inconsistencies across platforms caused many misses:

- `baldur's gate 3` vs. `baldurs gate 3`
- `black myth: wukong` vs. `black myth wukong`
- `elden ring` vs. `elden ring (video game)`

## Solution

- Remove parenthetical content (`...`)
- Remove punctuation, keep only letters, numbers, spaces
- Collapse multiple spaces, trim whitespace
- Extract main title before `:` as secondary match key
- Combine both match strategies with `unionByName`, deduplicate by game ID

# Insights

## Commercial Success Does Not Guarantee Female Audience Engagement :

- Of the 1,004 matched games, 348 (35%) scored high on commercial ratings but low on female engagement (combo 10)
- Nearly half of top-rated games failed to resonate with the female fandom community
- This confirms that the female gaming market remains systematically underserved despite its size

# Insights

## Female-Favored Games Generate Stronger Revenue Than Expected :

- Games with high female engagement but lower ratings (combo 01) averaged \$33.8M in estimated revenue: nearly double that of games with low engagement on both sides (combo 00, \$18.2M)
- Games that scored high on both dimensions (combo 11) had the highest average revenue at \$68.2M
- Female audience recognition correlates with stronger commercial returns, even when overall ratings are modest

# Insights

## Female Fans Show Deeper Emotional Investment in Underappreciated Games :

- Combo 01 games (commercially underrated but female-favored) had the highest average fanfiction word count at 9,342 words per fic
- This is ~26% longer than other groups, suggesting stronger emotional attachment
- Female players do not just engage more broadly. They engage more deeply with games that speak to them, indicating untapped creative and commercial potential

# Obstacles

- Normalizing engagement volume across games with vastly different player bases

Raw fanfiction counts and download numbers are not directly comparable

- No gender-disaggregated player data available

AO3 serves as a proxy for female audience engagement, but direct male/female split data from Steam does not exist

- Free-to-play games excluded from revenue analysis:

estimated revenue cannot be calculated via  $\text{price} \times \text{units sold}$ , and most publishers do not release financial breakdowns per title

# Summary

- Market Insights: The "Engagement-Revenue" Gap
- Big Data Implementation & Robustness
- Future Impact